



MX-SORT

Communication Interface Description

V1.1.0.0

MX-SORT

Communication Interface Description

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Rev. 1

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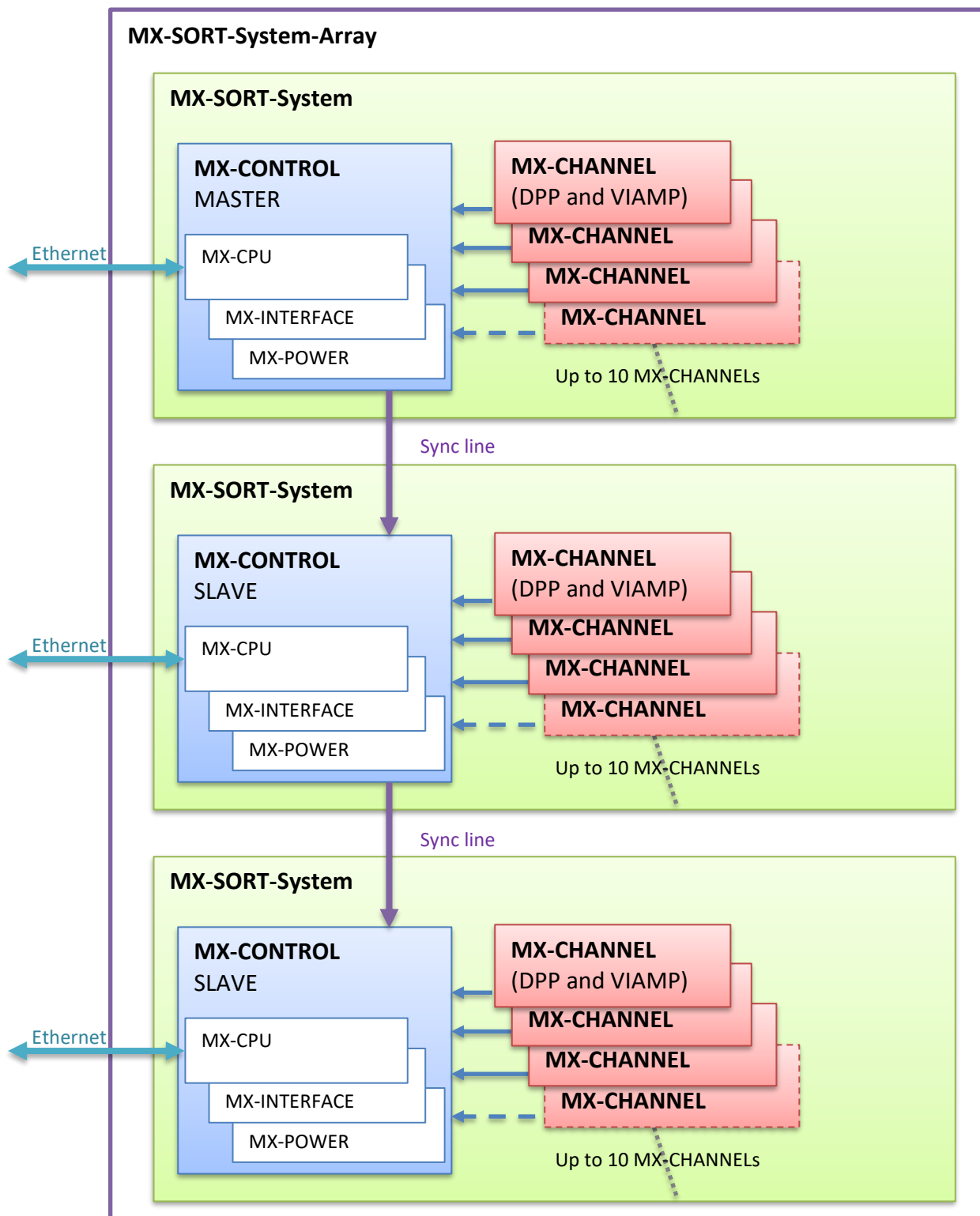
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1. Connection and System Overview

The MX-System consists of an MX-CONTROL and up to ten MX-CHANNELs (MX-DVKC-SPI combined with KETEK VIAMP-KC). Each MX-CHANNEL provides a spectral data channel. Several MX-Systems can be combined to an MX-System-Array to increase the channel count. Each MX-CONTROL provides synchronisation connectors to link several MX-Systems and to provide an MX-System-Array-wide channel read execution trigger.

The MX-CONTROL is equipped with an ethernet connection. This connection is used to transfer spectral data and to configure the system. The configuration is done via an HTTP RESTful API. There is also a web-based configuration implemented. The configuration is stored within the MX-CONTROL. After powering the system, the MX-CONTROL loads the configuration and applies it to all connected MX-CHANNELs. If an MX-CHANNEL is replaced, the settings of the previous MX-CHANNEL will be applied and spectral acquisition will be started automatically.

Each MX-CHANNEL's acquisition data is sent in a separate UDP packet.



2. Software

The MX-CONTROL Software is automatically started when the system is powered. After the boot procedure all channel configurations are loaded from the internal storage of the MX-CONTROL system and applied to each channel. After configuration each enabled MX-CHANNEL will switch to the run mode where spectral data is generated and send using UDP packets.

The MX-CONTROL software performs a synchronized readout of the spectra on each connected channel. The acquisition of spectra can be synchronized over several MX-CONTROLS. The spectra are serialized in the MX-CONTROL and sent as separate UDP packets per MX-CHANNEL to the destination host system. The UDP packet structure is described in the following chapter.

The MX-CONTROL is delivered with a webserver which allows the user to configure all MX-CONTROL and MX-CHANNEL settings using a web interface. All web interface settings are also parametrizable using a RESTful API for deeper integration in customer software.

2.1. UDP Packet Structure

Each active MX-CHANNEL produces one UDP packet per MX-CONTROL cycle. The UDP packets are structured in sections. Each section begins with the section length followed by the section type. The sections within a packet are always sorted ascending by their value. Currently the following section types are defined:

Name	Value
<invalid>	0
ChannelMetaData	1
SpectralMetaData	2
Spectrum	3
DiagnosticData	4
SingleChannelAnalyzerData	5

Due to transmission errors or EMC interference, it could happen that no spectral data is available. Depending on the channel settings the spectral meta data, the spectrum and the SCA data sections are left out in this packet or filled with default values. In both cases the "SpectralDataAvailable" bit within the UDP Meta data section is set to false.

In general, every data section starts with the length of the following section.

The interface is versioned with a major and minor number (e.g. 1.0). For future revisions of this interface we keep to the following guidelines regarding the version changes:

1. Adding of data elements to the end of the section leads to an increment of the minor interface version number.
2. New data elements are appended to the end of a section to maintain compatibility with older interface versions.
3. Adding a new data section type only increments the minor version number.
4. Structural changes which lead to incompatibility issues will require an increment of the major version number.

If you consider using the data length of the sections and the section type in your receiving code, you can maintain compatibility with newer minor interface versions even if you do not implement additional features of the newer MX-CONTROL software.

The content of the individual sections is presented in tabular form in the following sections.

2.1.1. Channel Meta Data Section

Start Byte	Data type	Name	Description
0	U16	SectionDataLen	Length of data section excluding SectionDataLen in bytes.
2	U8	SectionType	1: ChannelMetaData
3	U8	ArrIdx	MX-System Index as set in MX-CONTROL configuration
4	U8	ChnIdx	MX-Channel index of the packet within the MX-CONTROL (1 .. 10).
5	U8	IfcVerMajor	Communication interface version major number
6	U8	IfcVerMinor	Communication interface version minor number
7	U16	StatusBits	Status word Bit 0: DppAvailable Bit 1: SpectralDataAvailable Bit 2: SnapshotStatisticDataValid Bit 3: MxPowerSupplyState Bit 4: MxPowerSupplyStable Bit 5: SddTemperatureReady Bit 6-15: Spare
9	U8	ChannelState	0: Disabled 1: Uninitialized 2: WriteConfiguration 3: UpAndRunning 4: CommunicationReset 5: ReadDppConfiguration 6: Linetest 7: LinetestFailed 8: IncompatibleHardware
10	U16	CntCommunicationReset*	Counter incrementing each time the MX-CHANNEL communication is reset.
12	U16	CntUdpPacket*	This counter is incremented each time a UDP packet is sent. Consecutive UDP packet are identifiable with this counter.
14	U16	CntUdpSendErr*	This counter is incremented if an error occurred within the MX-CONTROL while sending the UDP data packet.
16	U16	CntMissedPacket*	This counter is increased if no valid data was received from an MX-CHANNEL.
18	U16	CntSyncErr*	This counter is increased if the received sync line data was invalid
20	U16	CntSnapshotStatInvalid*	This counter is increased if the snapshot run statistic data is invalid
22	Float	DppTemperature	Current temperature of the DPP. If not available the value is NaN
26	Float	SddTemperature	Current temperature of the SDD. If not available the value is NaN

*All counters are overflowing.



26(section1最后的
Intex)+4(最后一项的字节长度)

2.1.2. Spectral Meta Data Section

Start Byte	Data type	Name	Description
0	U16	SectionDataLen	Length of data section excluding SectionDataLen in bytes.
2	U8	SectionType	2: SpectralMetaData
3	U16	SpectCtr	Counter of the current MX-CONTROL cycle. This value is synchronised between MX-CONTROLS for MX-SORT-System-Arrays
5	Float	DppTransferTime	Transfer time of data between MX-CONTROL and MX-CHANNEL in seconds
9	Float	Realtime	Realtime in seconds
13	Float	Lifetime	Lifetime in seconds
17	U32	Events	Sum of events counted in spectrum
21	Float	InputCountRate	Input count rate in counts per second
25	Float	OutputCountRate	Output count rate in counts per second
29	Float	DeadtimeRatio	Deadtime as factor between 0.0 and 1.0
33	U8	BytesPerBin	Number of bytes per bin as configured
34	U16	NumberOfBins	Number of bins as configured
36	U16	StartBin	Offset in bins as configured

All counters are overflowing.

2.1.3. Spectrum Section

Start Byte	Data type	Name	Description
0	U16	SectionDataLen	Length of data section excluding SectionDataLen in bytes.
2	U8	SectionType	3: Spectrum
3	U8/U16 Array	Spectrum	Raw stream of spectral data (Number of Bytes = NumberOfBins * BytesPerBin)

All counters including spectral bins are overflowing.

2.1.4. Diagnostic Data Section

Start Byte	Data type	Name	Description
0	U16	SectionDataLen	Length of data section excluding SectionDataLen in bytes.
2	U8	SectionType	4: DiagnosticData
3	U16	StatusBits	Status word Bit 0: TemperatureWarning • MX-CPU Temperature > 85°C or • MX-CPU Upper Surface Temperature > 75°C or • MX-CPU Under Surface Temperature > 75°C or • MX-POWER Temperature > 75°C) Bit 1: TemperatureFault • MX-CPU Temperature > 95°C or • MX-CPU Upper Surface Temperature > 85°C or • MX-CPU Under Surface Temperature > 85°C or • MX-POWER Temperature > 85°C. If a temperature fault is detected processing of data is switched off to reduce the temperature. The bit is deactivated with a hysteresis of 5°C) Bit 2: MxPowerUndervoltageLockout (Input voltage < 15V, if this bit is triggered, all MX-CHANNELs are deactivated) Bit 3: MixedModeDetected (True if mixed mode for firmware is allowed and detected. If mixed mode is not allowed and detected all channels are switched to the state IncompatibleHardware) Bit 4-15: Spare
5	Float	MxCpuTemperature	MX-CPU Processor Temperature
9	Float	MxCpuUpperSurfaceTemperature	MX-CPU upper surface temperature
13	Float	MxCpuUnderSurfaceTemperature	MX-CPU under surface temperature
17	Float	MxCpuLoadCore1	MX-CPU processor load of first core
21	Float	MxCpuLoadCore2	MX-CPU processor load of second core
25	Float	MxPowerTemperature	MX-POWER temperature sensor reading

2.1.5. Single Channel Analyzer Data Section

Start Byte	Data type	Name	Description
0	U16	SectionDataLen	Length of data section excluding SectionDataLen in bytes.
2	U8	SectionType	5: SingleChannelAnalyzerData
3	U16	SCA 1 StartBin	Start bin of SCA
5	U16	SCA 1 Length	Number of bins summed up for current SCA
7	U32	SCA 1 Counts	Calculated Counts within SCA
11	U16	SCA 2 StartBin	continued
13	U16	SCA 2 Length	
15	U32	SCA 2 Counts	
19	U16	SCA 3 StartBin	
21	U16	SCA 3 Length	
23	U32	SCA 3 Counts	
27	U16	SCA 4 StartBin	
29	U16	SCA 4 Length	
31	U32	SCA 4 Counts	
35	U16	SCA 5 StartBin	
37	U16	SCA 5 Length	
39	U32	SCA 5 Counts	
43	U16	SCA 6 StartBin	
45	U16	SCA 6 Length	
47	U32	SCA 6 Counts	
51	U16	SCA 7 StartBin	
53	U16	SCA 7 Length	
55	U32	SCA 7 Counts	
59	U16	SCA 8 StartBin	
61	U16	SCA 8 Length	
63	U32	SCA 8 Counts	
67	U16	SCA 9 StartBin	
69	U16	SCA 9 Length	
71	U32	SCA 9 Counts	
75	U16	SCA 10 StartBin	
77	U16	SCA 10 Length	
79	U32	SCA 10 Counts	
83	U16	SCA 11 StartBin	
85	U16	SCA 11 Length	
87	U32	SCA 11 Counts	
91	U16	SCA 12 StartBin	
93	U16	SCA 12 Length	
95	U32	SCA 12 Counts	
99	U16	SCA 13 StartBin	
101	U16	SCA 13 Length	
103	U32	SCA 13 Counts	
107	U16	SCA 14 StartBin	
109	U16	SCA 14 Length	
111	U32	SCA 14 Counts	
115	U16	SCA 15 StartBin	
117	U16	SCA 15 Length	
119	U32	SCA 15 Counts	

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Start Byte	Data type	Name	Description
123	U16	SCA 16 StartBin	
125	U16	SCA 16 Length	
127	U32	SCA 16 Counts	

All counters including spectral bins are overflowing.

2.2. Web interface

The web interface of the MX-CONTROL can be reached under the set IP address. The default IP address is 172.28.110.100. It enables the user to change all MX-CONTROL and MX-CHANNEL settings. Additionally, it includes basic debugging as well as update features. The following chapters describe each menu entry.

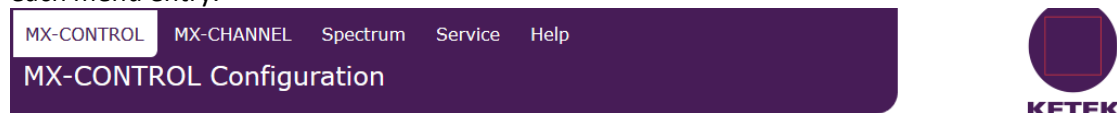


Figure 1: MX-CONTROL web interface menu

2.2.1. MX-CONTROL

The website MX-CONTROL shows some basic information about the controller as software and hardware version information and an overview of the current channel states. It is also used to configure all MX-CONTROL relevant settings.

Information

MX-CPU App Version: 1.1.0.0
MX-CPU UDP Protocol Version: 1.0
MX-CPU Serial Number: 1D9B145
MX-CPU MAC Address: 00:80:2F:22:B8:63
MX-CPU Linux Version: NI Linux Real-Time ARMv7-A 4.6.7-rt14-ni-5.2.0f0
MX-CPU Runtime Version: 17.0
MX-INTERFACE EEPROM State: Operating
MX-INTERFACE Hardware Revision: 1021_02_AA_01
MX-INTERFACE Item Number: MX-INTERFACE-SPI
MX-INTERFACE Serial Number: 9964704578
MX-INTERFACE Manufacturing Date: 2019-09-05
MX-POWER EEPROM State: Operating
MX-POWER Hardware Revision: 1022_02_AA_01
MX-POWER Item Number: MX-POWER-10
MX-POWER Serial Number: 9949386863
MX-POWER Manufacturing Date: 2019-09-05
MX-CPU StatusBits0: 0000
MX-CPU Temperature: 53.625 [°C]
MX-CPU Upper Surface Temperature: 47.0625 [°C]
MX-CPU Under Surface Temperature: 48.6875 [°C]
MX-CPU Load Core 1: 9.182408 [%]
MX-CPU Load Core 2: 48.76275 [%]
MX-POWER Temperature: 32.5625 [°C]

Settings

IP Address:
Subnet Mask:
UDP Destination Host:
Cycle Time: [µs]
Line Test Time: [µs]
MX-SYSTEM-Index:
Sync Mode:
Allow Mixed Firmware:

Apply Refresh

Figure 2: MX-CONTROL Information and Settings section

To change a specific setting, it must be edited in the input field in the Settings section of the website. To apply the new setting the button "Apply" must be pushed. To reload the site and reset the input

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fields to the current settings you can push the button "Refresh". The following input fields are editable:

Input field	Default	Description
IP Address	172.28.110.100	Static IPV4 address of the MX-CONTROL.
Subnet Mask	255.255.255.0	Subnet mask of the MX-CONTROL
UDP Destination Host	172.28.110.110:27001	Target IP and port for the UDP data
Cycle Time	10000	System cycle time in μ s
Line Test Time	50000	Duration of MX-CHANNEL line test in μ s. The line test is used to check the basic communication capability of the MX-CONTROL with the connected DPPs. It is not recommended to change this setting.
MX-SYSTEM-Index	0	Index of the MX-SORT-System within an MX-SORT-System-Array.
Sync Mode	Internal Trigger	Controls the sync line behaviour. Possible selections are: - Internal Trigger (self-triggering) - React on external trigger (Slave) - Generate external trigger (Master)
Allow Mixed Firmware	False	Controls the behaviour if the system detects different firmware versions in the connected MX-CHANNELs. It is possible to attach different MX-CHANNEL versions on a system but it is not recommended as it influences the timing behaviour of the system according the synchronicity of the received channel data.

At the end of the site there is an overview of the current states of all channels. All error counters from the channel meta data section are also shown in this table. It is possible to reset all counter values with the button "Reset Counters". There is also a "Refresh" button that reloads the website.

Channel States								
Chn	State	StatusBits	ComReset	UDP	UDP Err	Missed	Sync Err	Snap Err
1	UpAndRunning	111111	0	0	14042	0	0	0
2	UpAndRunning	111111	0	0	14042	0	0	0
3	UpAndRunning	111111	0	0	14042	0	0	0
4	UpAndRunning	111111	0	0	14042	0	0	0
5	UpAndRunning	111111	0	0	14042	0	0	0
6	UpAndRunning	111111	0	0	14042	0	0	0
7	UpAndRunning	111111	0	0	14042	0	0	0
8	UpAndRunning	111111	0	0	14042	0	0	0
9	UpAndRunning	111111	0	0	14042	0	0	0
10	UpAndRunning	111111	0	0	14042	0	0	0

Reset Counters
Refresh

Figure 3: MX-CONTROL Channel state overview

2.2.2. MX-CHANNEL

The website MX-CHANNEL shows some basic DPP information about the channel as software and hardware version information. It is also used to configure all MX-Channel relevant settings.

Information	
DPP Serial Number:	n/a
DPP Version:	PIC -0, DSP -0, FIPPI -0

Settings	
MX-CHANNEL:	Channel 1 ⓘ
Enable Channel:	Disabled ☐
Send Spectral Meta Data:	Always send ☒
Send Spectrum:	Always send ☒
Send Diagnostic Data:	Always send ☒
Send SCA Data:	Do not send ☒

Genset	
Gain:	1 ☐
Switched Gain:	high ☒
Bytes Per Bin:	2 ☒
Number Of Bins:	2048 ☒
Start Bin:	0 ☒
Decimation Factor:	2 ☒

Parset	
Peaking Time:	0.1 ☐
Fine Gain Trim:	1 ☐
Gap Time:	0,3 ☒ [μs]
Pileup Inspection Interval Offset:	0 ☒ [μs]
Peak Capture Mode:	Peak Sensing ☒
Peak Sampling Time Offset:	-0,2 ☒ [μs]
Fast Trigger Peaking Time:	0,05 ☒ [μs]
Fast Trigger Gap Time:	0 ☒ [μs]
Fast Trigger Max Width:	0,4 ☒ [μs]
Trigger Threshold:	30 ☒
Baseline Threshold:	0 ☒
Energy Threshold:	0 ☒
Baseline Average Length:	512 ☒

☐ to all channels"/>

Figure 4: MX-CHANNEL Information and Settings section

To change a setting, it must be edited in the input field in the Settings section of the website. To apply the new setting the button "Apply" must be pushed. At the end of each input field line there is a checkbox. If the button "Apply ☒ to all channels" is pushed the settings of the input fields with a checked checkbox are applied to all MX-CHANNELS of the MX-CONTROL. To reload the site and reset the input fields to the current settings you can push the button "Refresh".

The settings for single channel analysis (SCA) are hidden until "Send SCA Data" is set to "Always send" or "Send if available". Up to 16 regions of interest are configurable by defining the start bin and the length of the region. If the SCA evaluation is activated the Start Bin setting for the spectrum is forced to 0. The calculation is executed within the MX-CONTROL. SCA data is only configurable in the region transferred to the MX-CONTROL system.

SCA Settings		
SCA	Start Bin	Length
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	0	0
10	0	0
11	0	0
12	0	0
13	0	0
14	0	0
15	0	0
16	0	0

Figure 5:MX-CHANNEL SCA Settings

The following input fields are editable:

Element	Default	Description
MX-CHANNEL	Channel 1	Channel selector to switch the currently edited channel (Channel 1 .. Channel 10)
Enable Channel	Disabled	Enables or disables the channel. - Disabled - Enabled
Send Spectral Meta Data	Always send	Enables or disables the sending of spectral meta data. - Do not send - Always send - Send if available
Send Spectrum	Always send	Enables or disables the sending of the spectrum. - Do not send - Always send - Send if available
Send Diagnostic Data	Always send	Enables or disables the sending of diagnostic data. - Do not send - Always send
Send SCA Data	Do not send	Enables or disables the sending of single channel analyser data. - Do not send - Always send - Send if available
Gain	1.0	Gain value of the DPP. Used to calibrate the system. Value is coerced between 0.25 and 4.0
Switched Gain	high	Analog gain of the DPP - low - high
Bytes Per Bin	2	Number of bytes per bin (1 or 2).

Element	Default	Description
Number Of Bins	2048	Number of bins in each spectrum. Maximum is 4096 bins.
Start Bin	0	Offset in bins. Can be changed in steps of 2. If Send SCA Data is active this value is forced to 0.
Decimation Factor	2	Number of bins combined within the DPP.
Peaking Time	0.1	Active peaking time in μ s
Fine Gain Trim	1.0	Value between 0.5 and 2.0 to change the fine gain to adapt to a specific peaking time.
Gap Time	0.3	Gap time in μ s used for energy filtering. Can be changed in steps of 0.025 μ s.
Pileup Inspection Interval Offset	0.0	Pileup inspection interval offset. Can be changed in steps of 0.025 μ s.
Peak Capture Mode	Peak Sensing	Peak capturing mode. Possible values are: • Peak Sensing • Peak Sampling
Peak Sampling Time Offset	-0.2	Peak sampling time offset in μ s. Only used if "Peak Capture Mode" is set to "Peak Sampling". Can be changed in steps of 0.025 μ s.
Fast Trigger Peaking Time	0.05	Peaking time setting of fast trigger filter in μ s. Can be changed in steps of 0.025 μ s.
Fast Trigger Gap Time	0.0	Gap time setting of fast trigger filter in μ s. Can be changed in steps of 0.025 μ s.
Fast Trigger Max Width	0.4	Max width setting of fast trigger filter in μ s. Can be changed in steps of 0.025 μ s.
Trigger Threshold	30	Trigger threshold (0 .. 4095)
Baseline Threshold	0	Baseline threshold (0 .. 4095)
Energy Threshold	0	Energy threshold (0 .. 4095)
Baseline Average Length	512	Baseline average length between 1 and 512.
SCA 1..16 Start Bin	0	Only accessible if Send SCA Data is set to "Always send" or "Send if available". Start bin for SCA analyzer.
SCA 1..16 Length	0	Only accessible if "Send SCA Data" is set to "Always send" or "Send if available". Number of Bins for SCA analyzer.

2.2.3. Spectrum

The website spectrum is for debugging purposes only. It shows the currently stored spectrum when the site was loaded. To update the view the site needs to be refreshed. The spectral data for this debugging view is only updated every 3 seconds.

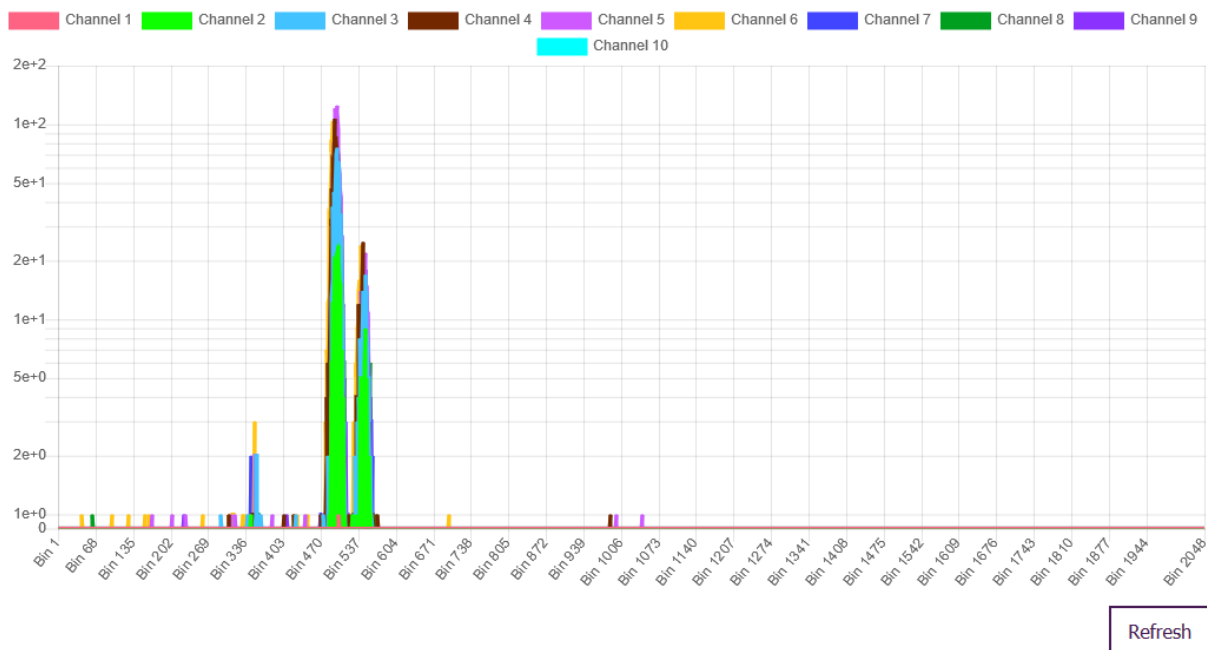


Figure 6: Spectrum Website

2.2.4. Service

The service website enables the user to update or downgrade the system to a different firmware version. To change the firmware the new firmware file needs to be selected. The firmware file format is a single Zip-Folder containing all necessary software components. After selecting the new firmware, the button Update needs to be pushed to start the procedure. During the update procedure the system is rebooted. The MX-CONTROL and MX-CHANNEL settings are not changed by the update procedure.

There is also a button to manually induce a reboot of the system.

To easily copy or backup channel settings in a MX-SORT-System-Array there is the possibility to download the current channel configuration of all MX-CHANNELS. This config files can also be uploaded to the system.

Update

MX-CPU App Version: 1.1.0.0
MX-CPU UDP Protocol Version: 1.0
MX-CPU Linux Version: NI Linux Real-Time ARMv7-A 4.6.7-rt14-ni-5.2.0f0
MX-CPU Runtime Version: 17.0
Select firmware file: Keine Datei ausgewählt.
Update Result:

Channel configuration

Select configuration file: Keine Datei ausgewählt.
Config Result:

Figure 7: Service Website

2.2.5. Help

The help webpage shows additional information like the vendor information and used third party licences.

Contact

KETEK GmbH
Creative Detector Solutions
www.ketek.net

Third party licences

This appendix contains licensing information about third-party products included with MX-SORT:

- ▶ JDP Science Common Utilities v1.1.1.7 by JDP Science
- ▶ JSONtext v1.2.10.74 by JDP Science
- ▶ NI Linux RT Utilities v1.1.1.9 by National Instruments
- ▶ NI Linux RT Errno v1.1.2.9 by National Instruments
- ▶ NI Linux RT Inter-Process Communication v1.5.1.19 by National Instruments
- ▶ jQuery v3.3.1
- ▶ Chart.js v2.7.2

Figure 8: Help Website

2.3. RESTful API

The MX-CONTROL provides webservises to configure all MX-CONTROL and MX-CHANNEL settings. The services are listening on port 8000 of each MX-CONTROL. The exchange data format is JSON. Each configuration URL can be interfaced using a GET request to retrieve the current settings and a PUT request to modify the settings. GET and PUT requests have the same response data format. PUT requests should contain JSON data with the new settings. It is possible to only send a subset of the settable items. Not all data is modifiable e.g. serial numbers and software versions of the connected hardware. Those items are combined in a JSON object named "Information". If non-modifiable items are sent in a PUT request those items are ignored without an error message. All other items are checked against limits and coerced during command processing. The response JSON data contains the complete set of settings and information. There are also two command PUT only requests (Reboot and ResetCounters) implemented, which have no JSON data.

2.3.1. Controller (GET/PUT)

Command URL: `http://<xxx.xxx.xxx.xxx>:8000/Configuration/Controller`

Example: `"http://172.28.110.100:8000/Configuration/Controller"`

JSON Example:

```
{
  "Settings":{
    "IpAddress":"172.28.110.100",
    "SubnetMask":"255.255.255.0",
    "UdpDestinationHost":"172.28.110.110:27001",
    "CycleTime":10000,
    "LineTestTime":50000,
    "MxSystemIndex":0,
    "SyncMode":0,
    "AllowMixedFirmware":0
  },
  "Information":{
    "MxCpuAppVersion": "1.1.0.0",
    "MxCpuUdpProtocolVersion": "1.0",
    "MxCpuSerialNumber": "1D5B97F",
    "MxCpuMacAddress": "00:80:2F:22:26:D7",
    "MxCpuLinuxVersion": "NI Linux Real-Time ARMv7-A 4.6.7-rt14-ni-5.2.0f0",
    "MxCpuRuntimeVersion": "17.0",
    "MxInterfaceEepromState": "Operating",
    "MxInterfaceHardwareRevision": "1021_02_AA_01",
    "MxInterfaceItemNumber": "MX-INTERFACE-SPI",
    "MxInterfaceSerialNumber": "MXI0018",
    "MxInterfaceManufacturingDate": "2019-08-02",
    "MxPowerEepromState": "Operating",
    "MxPowerHardwareRevision": "1022_02_AA_01",
    "MxPowerItemNumber": "MX-POWER-10",
    "MxPowerSerialNumber": "MXP0018",
    "MxPowerManufacturingDate": "2019-08-02",
    "MxCpuStatusBits": 0,
    "MxCpuTemperature": 54.875,
    "MxCpuUpperSurfaceTemperature": 46,
    "MxCpuUnderSurfaceTemperature": 47.3125,
    "MxCpuLoadCore1": 7.556575,
    "MxCpuLoadCore2": 35.9895,
    "MxPowerTemperature": 41.875
  }
}
```

Settings

Element	Default	Description
IpAddress	172.28.110.100	Static IPV4 address of the MX-CONTROL.
SubnetMask	255.255.255.0	Subnet mask of the MX-CONTROL
UdpDestinationHost	172.28.110.110:27001	Target IP and port of the UDP data
CycleTime	10000	System cycle time in μ s
LineTestTime	50000	Duration of MX-CHANNEL line test in μ s. The line test is used to check the basic communication capability of the MX-CONTROL with the connected DPPs.
MxSystemIndex	0	Index of the MX-SORT-System within an MX-SORT-System-Array.
SyncMode	0	Controls the sync line behaviour. 0: Internal Trigger (self-triggering) 1: React on external trigger (Slave) 2: Generate external trigger (Master)
AllowMixedFirmware	0	Controls the behaviour if the system detects different firmware versions in the connected MX-CHANNELs. It is possible to attach different MX-CHANNEL versions on a system but it is not recommended as it influences the timing behaviour of the system according the synchronicity of the received channel data. 0: do not allow mixed firmware 1: allow mixed firmware

Information

Element	Description
MxCpuAppVersion	Software Version of the MX-CONTROL software.
MxCpuUdpProtocolVersion	Version of the active MX-CONTROL UDP protocol.
MxCpuSerialNumber	Serial number of the MX-CONTROL.
MxCpuMacAddress	MAC address of the MX-CONTROL Ethernet port.
MxCpuLinuxVersion	Linux Version of MX-CONTROL
MxCpuRuntimeVersion	Runtime Version for MX-CONTROL software
MxInterfaceEepromState	MX-INTERFACE EEPROM status
MxInterfaceHardwareRevision	MX- INTERFACE hardware revision
MxInterfaceItemNumber	MX- INTERFACE item number
MxInterfaceSerialNumber	MX- INTERFACE serial number
MxInterfaceManufacturingDate	MX- INTERFACE manufacturing date
MxPowerEepromState	MX-POWER EEPROM status
MxPowerHardwareRevision	MX- POWER hardware revision
MxPowerItemNumber	MX- POWER item number
MxPowerSerialNumber	MX- POWER serial number
MxPowerManufacturingDate	MX- POWER manufacturing date

Element	Description
MxCpuStatusBits	Status word Bit 0: TemperatureWarning • MX-CPU Temperature > 85°C or • MX-CPU Upper Surface Temperature > 75°C or • MX-CPU Under Surface Temperature > 75°C or • MX-POWER Temperature > 75°C) Bit 1: TemperatureFault • MX-CPU Temperature > 95°C or • MX-CPU Upper Surface Temperature > 85°C or • MX-CPU Under Surface Temperature > 85°C or • MX-POWER Temperature > 85°C. If a temperature fault is detected processing of data is switched off to reduce the temperature. The bit is deactivated with a hysteresis of 5°C) Bit 2: MxPowerUndervoltageLockout (Input voltage < 15V, if this bit is triggered, all MX-CHANNELs are deactivated) Bit 3: MixedModeDetected (True if mixed mode for firmware is allowed and detected. If mixed mode is not allowed and detected all channels are switched to the state IncompatibleHardware) Bit 4-15: Spare
MxCpuTemperature	MX-CPU Processor Temperature
MxCpuUpperSurfaceTemperature	MX-CPU upper surface temperature
MxCpuUnderSurfaceTemperature	MX-CPU under surface temperature
MxCpuLoadCore1	MX-CPU processor load of first core
MxCpuLoadCore2	MX-CPU processor load of second core
MxPowerTemperature	MX-POWER temperature sensor reading

Command URL: `http://<xxx.xxx.xxx.xxx>:8000/Configuration/Channel<1..10>`

JSON Example:

Changing the Channel settings always induces the reconfiguration of the specific MX-CHANNEL. Due to technical reasons some of the settings are bound to a specific peaking time. Those peaking time specific parameters are structured in a JSON object named "Parset" within the "Settings" object. The MX-CONTROL internally stores one "Parset" per peaking time. Only one "Parset" is active at a time. The following rules apply to the handling of "Parset" data and peaking time settings:

1. Only the active peaking time's "Parset" data is returned by a Channel GET/PUT command.
2. If a PUT command is sent without a new "PeakingTime" setting all received "Parset" parameters are applied to the currently active peaking time.
3. If a PUT command containing a new "PeakingTime" setting is sent, the peaking time parameter is always changed first, independent of the order within the JSON data. All received "Parset" parameters are then applied to the new peaking time.

Settings

Element	Default	Description
EnableChannel	0	Enables or disables the channel. 0: Disabled 1: Enabled
SendSpectralMetaData	1	Enables or disables the sending of spectral meta data. 0: Do not send 1: Always send 2: Send if available
SendSpectrum	1	Enables or disables the sending of the spectrum. 0: Do not send 1: Always send 2: Send if available
SendDiagnosticData	1	Enables or disables the sending of diagnostic data. 0: Do not send 1: Always send
SendSingleChannelAnalyzerData	0	Enables or disables the sending of single channel analyser data. 0: Do not send 1: Always send 2: Send if available
Gain	1.0	Gain value of DPP. Used to calibrate system. Value is coerced between 0.25 and 4 (tbd)
SwitchedGain	1	Analog switch gain of the DPP. 0: low 1: high
BytesPerBin	2	Number of bytes per bin (1 or 2).
NumberOfBins	2048	Number of bins in each spectrum. Maximum is 4096 bins.
StartBin	0	Offset in bins. Can be changed in steps of 2.
DecimationFactor	2	Number of bins combined within the DPP.
PeakingTime	0.1	Active peaking time in μ s
Parset		JSON Object containing all peaking time dependent items. Structure described in table below.
SingleChannelAnalyzerConfig		Array of 16 Objects of type SingleChannelAnalyzerConfigElement. This is the configuration for the SCA mode.

Parset

Element	Default	Description
FineGainTrim	1.0	Value between 0.5 and 2.0 to change the fine gain to adapt to a specific peaking time.
GapTime	0.3	Gap time in μs used for energy filtering. Can be changed in steps of $0.025\mu\text{s}$.
PileupInspectionIntervalOffset	0.0	Pileup inspection interval offset. Can be changed in steps of $0.025\mu\text{s}$.
PeakCaptureMode	0	Peak capturing mode. Possible values are: 0: Peak Sensing 1: Peak Sampling
PeakSamplingTimeOffset	-0.2	Peak sampling time offset in μs . Only used if PeakCaptureMode is set to "Peak Sampling". Can be changed in steps of $0.025\mu\text{s}$.
FastTriggerPeakingTime	0.05	Peaking time setting of fast trigger filter in μs . Can be changed in steps of $0.025\mu\text{s}$.
FastTriggerGapTime	0.0	Gap time setting of fast trigger filter in μs . Can be changed in steps of $0.025\mu\text{s}$.
FastTriggerMaxWidth	0.4	Max width setting of fast trigger filter in μs . Can be changed in steps of $0.025\mu\text{s}$.
TriggerThreshold	30	Trigger threshold (0 .. 4095)
BaselineThreshold	0	Baseline threshold (0 .. 4095)
EnergyThreshold	0	Energy threshold (0 .. 4095)
BaselineAverageLength	512	Baseline average length between 1 and 512.

SingleChannelAnalyzerConfigElement

Element	Default	Description
StartBin	0	Start bin for the channel
Length	0	Number of bins to add for the channel including the start bin.

Information

Element	Description
PeakingTimes	Array of peaking times settable for this channel in μs
DppSerialNumber	DPP serial number of the channel
DppVersion	DPP version string of the channel

2.3.3. Reboot (PUT)

Command URL: `http://<xxx.xxx.xxx.xxx>:8000/Configuration/Reboot`

Example: " `http://172.28.110.100:8000/Configuration/Reboot`"

JSON Structure: none

Put only command to trigger reboot of MX-CONTROL.

2.3.4. ResetCounters (PUT)

Command URL: `http://<xxx.xxx.xxx.xxx>:8000/Configuration/ResetCounters`

Example: " `http://172.28.110.100:8000/Configuration/ResetCounters`"

JSON Structure: none

Put only command to trigger reset of MX-CONTROL counter values. The counters are sent within the UDP ChannelMetaData described in a previous chapter.

2.4. Timing considerations

The MX-CONTROL system runs on a 2-core processor. Processor 2 is exclusively used for UDP packet processing. The processor load on core 2 is an indicator for the current system load state. All other tasks are executed on core 1 of the system. During the configuration of a channel sending of UDP packets is omitted. The channel state in the web interface is updated during configuration.

It is not recommended to make excessive use of the REST interface or the webserver during the evaluation of UDP packets, as the web server and web service execution can influence the timing behaviour and lead to missed packets.

The main limiting factor of the system is the transfer of the data between the MX-CHANNEL and the MX-CONTROL using SPI. The transfer is executed in parallel for all attached channels and therefore the maximum transfer time of all channels is important for the timing. It is necessary to ensure that the cycle time of the system is set to a value greater than the biggest DPP transfer time to ensure system stability.

The DPP transfer time depends on the bytes per bin and the number of bins transferred per cycle. Currently we are testing the system with the following timing settings:

CycleTime	Bytes per bin	Number of bins
10000µs	2	2048
5000µs	2	1024

The system settings are not limited to those timing values but these settings are recommended stable timing scenarios. It is possible to configure unstable timing settings but the webserver is always reactive to switch back to a stable state. Sometimes a reboot of the system is necessary. The number of channels attached to the system has also an influence on the timing, as the serialisation of the data has to be executed on the MX-CONTROL system. Therefore it is possible to increase the system performance by using less MX-CHANNELs per MX-CONTROL.

3. FAQ / Troubleshooting

UDP packet loss at receiving site

We recommend to connect only one MX-CONTROL to one receiving ethernet port. If you experience packet loss in this configuration you can increase the input buffer length within your network driver setting to improve communication stability.

4. Contact

If you have any questions regarding this or any of our products please do not hesitate to contact us via e-mail or phone.

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5. Glossary

Term	Description
API	Application Programming Interface
Big-endian	High byte first
DPP	Digital Pulse Processor
FAQ	Frequently Asked Questions
Float	Single Precision Floating Point Number (32 Bit, IEEE 754-1985, big-endian)
HTTP	Hypertext Transfer Protocol
IP	Internet Protocol
JSON	Java Script Object Notation
Little-endian	Low byte first
Recursion	See Recursion
REST	Representational State Transfer
SDD	Silicon Drift Detector
U16	Unsigned integer with 16 bits of storage, big-endian
U32	Unsigned integer with 32 bits of storage, big-endian
U8	Unsigned integer with 8 bits of storage
UDP	User Datagram Protocol
URL	Uniform Resource Locator